Chapter 4

Demand, Supply, and Equilibrium

ECON 003
UC Riverside · Summer 2026



Where We Left Off

Chapter 3 handed you a price and asked what to do about it.
Chapter 4 explains where the price came from.

CHAPTER 3 TOOK PRICES AS GIVEN

Rent was $1,180 for an apartment, rent decreased as we moved farther away from downtown.

BUT PRICES ARE NOT GIVEN

Somebody set that rent. If it is set too high, it takes longer to find a tenant.

THAT IS THIS CHAPTER

Buyers, sellers, and the one price that makes their plans fit together.

Chapter 3 parked one question twice: why does an apartment near the center cost more? Today you get the machinery to answer it.

Today's Roadmap

01 **What a Market Is**

The definition, and the strange conditions that make it "perfectly competitive"

02 **How Buyers Behave**

The demand curve — and the one distinction that costs the most marks

03 **How Sellers Behave**

The supply curve — the same logic, running the other way

04 **Putting Them Together**

Equilibrium, shortages, surpluses, and how to work any shift problem

05 **When Price Is Not Free to Move**

Price ceilings, campus parking, and the evidence on gasoline

PART 1

What a Market Is

Market Definition

A market is a group of economic agents trading a good or service, plus the rules and arrangements for trading it.

WHO



Buyers and sellers. Households, firms, governments — anyone on either side of the transaction.

WHAT



One good or service at a time. The market for gasoline is not the market for oil, and neither is the market for cars.

HOW



Who may trade, how prices are posted, how disputes settle. An auction, a shop, an app — all different rules.

A Market Does Not Need a Place

Some markets have an address. Most of the ones you use do not.



A market you can stand in

A PHYSICAL PLACE

The farmers' market on University Avenue. The Target checkout. The NYSE floor.

NO PLACE AT ALL

eBay. Uber. The market for used textbooks in a group chat. Buyers and sellers never occupy the same room.

The market for oil has no building. It is a price that thousands of people around the world are simultaneously reacting to.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Perfectly Competitive

In a **perfectly competitive market**: every seller sells an identical good, no single buyer or seller can influence the price, and everyone trades at the same price.

IDENTICAL GOODS

One barrel of West Texas crude is the same as any other.
Nobody asks whose it was.

Crude oil, wheat, sugar, foreign currency — genuinely identical. Coffee shops and sneakers are not.

NOBODY IS BIG ENOUGH

You buying or not buying gasoline this week does not move the price of gasoline.

You are one of roughly three hundred million American buyers. Your decision is invisible in the price.



One posted price, for all

Almost no real market satisfies all three conditions exactly — and the model still predicts well.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Price

The **market price** is the price at which buyers and sellers conduct transactions.
The **competitive equilibrium price** equates the quantity demanded and the quantity supplied.

A price is a selection device. It picks out the sellers who can produce cheaply and the buyers who value the good highly — and lets exactly those two groups trade.

ON THE SELLERS' SIDE

At $50 a barrel, a producer whose cost is $30 sells. A producer whose cost is $70 does not.

The market price sorted producers by their efficiency, and the low-cost ones got through.

ON THE BUYERS' SIDE

At $50, the buyer who values a barrel at $80 buys. The buyer who values it at $40 walks.

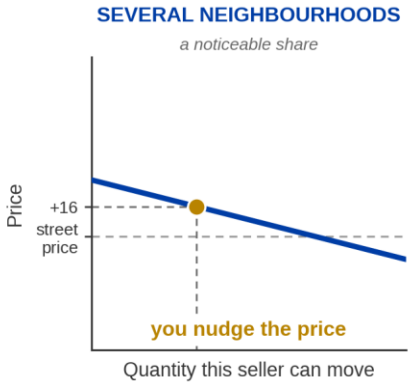
The people who wanted it most are the people who ended up with it.

Price Taker to Price Maker

Same product, same buyers, same city. The only thing that moved is how much of the market one seller controls.











PRICE TAKER

Buyers have an identical alternative one corner away. Raise your price and every buyer walks.

IN BETWEEN

Buyers have alternatives much farther away so they may tolerate elevated price to some extent.

PRICE MAKER

The curve you face IS the market curve. Tweak the price to extract max values from customers.

A price taker faces a flat demand curve; a price maker faces a sloped one. The whole difference is whether buyers have somewhere else to go.

PART 2

How Buyers Behave

Demand vs Quantity Demanded

QUANTITY DEMANDED

The amount of a good buyers are willing to purchase at a given price.

DEMAND SCHEDULE

A table listing the quantity demanded at each price, holding all else equal.

The same information as a list.

DEMAND CURVE

The same table, plotted. Price vertical, quantity horizontal.

The same information as a picture.

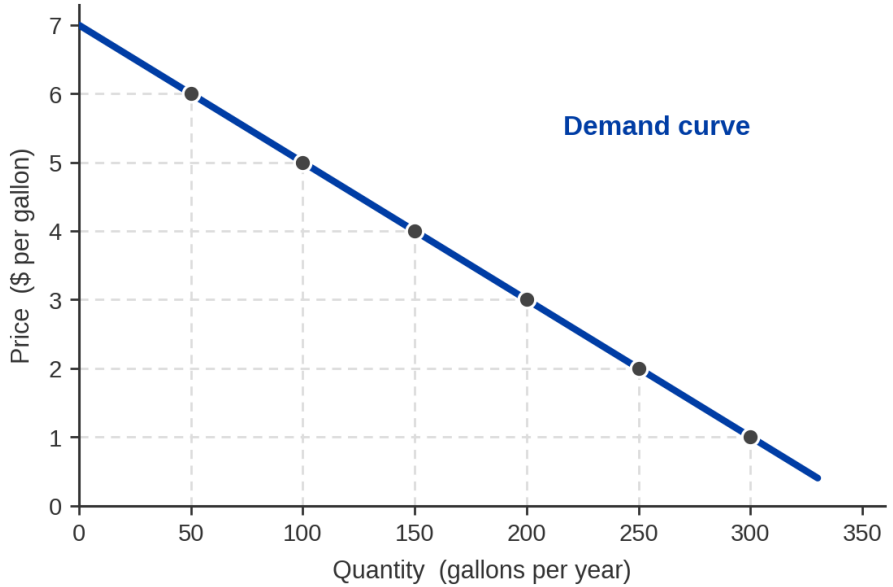


How much would you pay for this?

Everyone writes down a different number — and every one of those numbers is a point on that person's demand curve. And it represents the price at which your quantity demanded equals 1.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Exhibit 4.1 — One Buyer's Demand for Gasoline



CHLOE'S DEMAND SCHEDULE

Price ($/gal)	Quantity demanded (gallons/year)
$6	50
$5	100
$4	150
$3	200
$2	250
$1	300

Why Does It Slope Down?

Because the value of one more unit falls as you get more of them — which is the marginal reasoning from Chapter 3.

THE SECOND SLICE

The first slice is worth a lot. The second is worth less. By the fourth you are full.

Same for gallons.

SO PRICE HAS TO FALL

You will only buy the fourth gallon if it costs less than the fourth gallon is worth to you.

Cheaper price, more gallons.

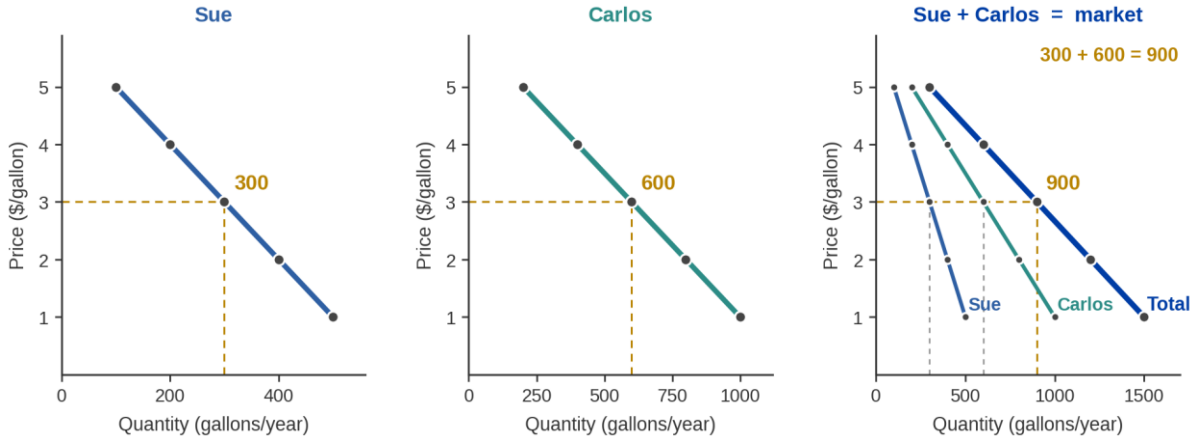
AND NEW BUYERS ARRIVE

In a market curve there is a second reason: at a lower price, people who were priced out entirely start buying at all.

Remember as we optimize, we compare marginal benefits vs marginal costs.

Exhibit 4.2 — Adding Buyers Together

The market demand curve is the sum of every individual buyer's demand — added across, at each price.



at $3 a gallon: Sue's 300 + Carlos's 600 = the market's 900

Add horizontally, never vertically. You are adding quantities at a common price — not prices at a common quantity.

The Law of Demand

In almost all cases, the quantity demanded rises when the price falls, ceteris paribus — holding all else equal.

“IN ALMOST ALL CASES”

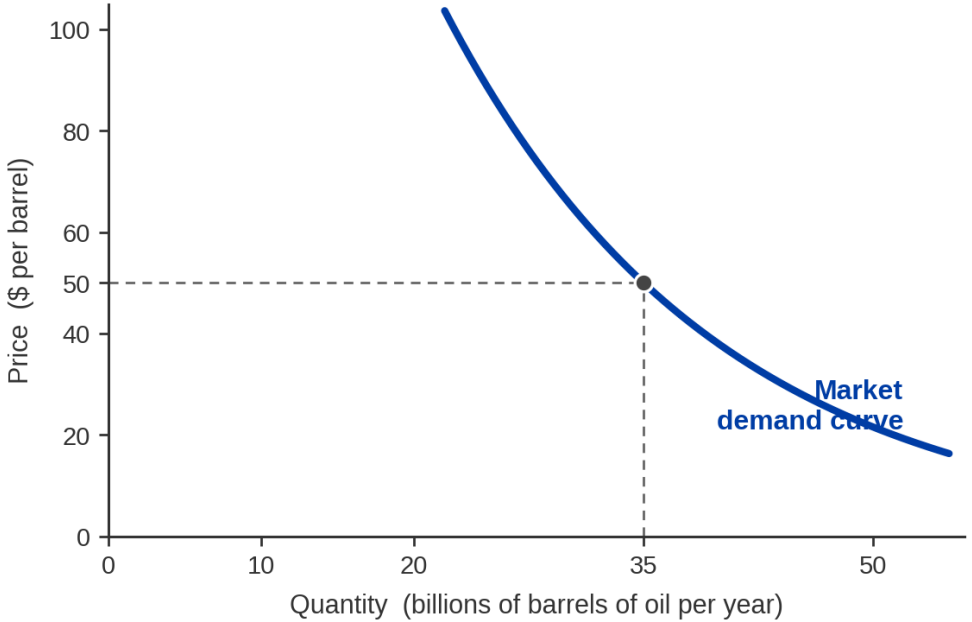
There are rare goods where demand rises with price — you will meet them in a later course.

For this course, and for essentially every good you will be asked about: down-sloping.

CETERIS PARIBUS

Income, tastes, other prices, expectations — all frozen. Only the good's own price moves.

Exhibit 4.3 — The Same Thing, for a Whole Country



SAME CURVE, NEW UNITS

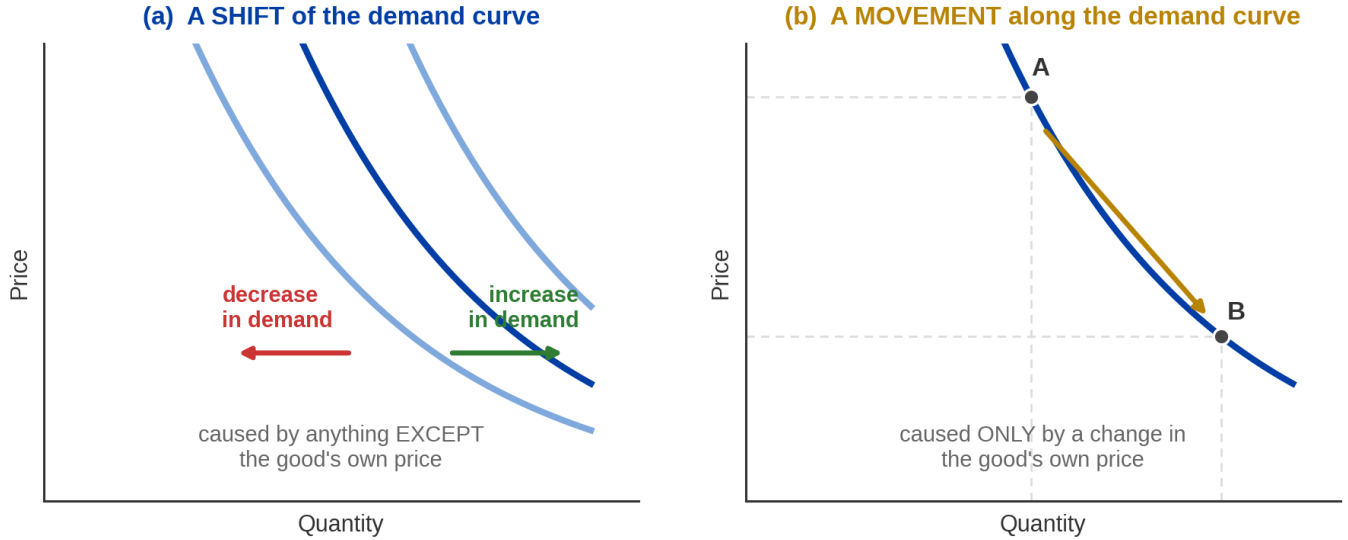
Chloe: gallons a year,
one person.

This: BILLIONS of barrels
a year, everybody in the US.

At $50 a barrel the world
wants about 35 billion barrels a year.

Illustrative figures from Acemoglu, Laibson & List, Microeconomics 3e — not today's market. Brent was about $88 a barrel on 31 July 2026.

Exhibit 4.4 — The Distinction of Shifts vs Movements



A SHIFT — the whole curve moves

Caused by ANYTHING EXCEPT the good's own price.
We now demand a different amount at every price.

A MOVEMENT — you slide along it

Caused ONLY by a change in the good's own price.
The curve itself never moved.

Five Things That Shift Demand



TASTES & PREFERENCES

What you like changes



INCOME & WEALTH

What you can afford changes



PRICES OF RELATED GOODS

Substitutes and complements



NUMBER OF BUYERS

More buyers, more demand



EXPECTATIONS ABOUT THE FUTURE

What you think happens next

None of these is the price of the good itself.

Income: Normal Goods and Inferior Goods

When your income rises, demand for a normal good rises — and demand for an inferior good falls.

NORMAL GOODS — most things

Income up → demand shifts RIGHT.

Restaurant meals. Air travel.
New clothes. You buy more of them when you can afford to.

This is the ordinary case.

INFERIOR GOODS — the odd ones

Income up → demand shifts LEFT.

Not bad goods — goods you buy because the better alternative was out of reach.



Bought less as incomes rise

Image: Acemoglu, Laibson & List, Microeconomics 3e

Related Goods: Substitutes and Complements



SUBSTITUTES — used INSTEAD of each other

Price of one goes UP → demand for the other goes UP.
Coffee and tea. Uber and Lyft. Chicken and beef.

COMPLEMENTS — used TOGETHER

Price of one goes UP → demand for the other goes DOWN.
Printers and ink. Phones and cases. Cars and gasoline.

Gasoline gets expensive → demand for large SUVs falls → demand for hybrids rises

One price change, two curves moved in opposite directions. Both of those are shifts, not movements along the demand curves.

Image: Acemoglu, Laibson & List, Microeconomics 3e

PART 3

How Sellers Behave

Sellers' Version of the Story

QUANTITY SUPPLIED

The amount of a good sellers are willing to sell at a given price.

SUPPLY SCHEDULE

A table listing the quantity supplied at each price, holding all else equal.

SUPPLY CURVE

The same table, plotted. Same axes as before — price vertical, quantity horizontal.

This one slopes UP.

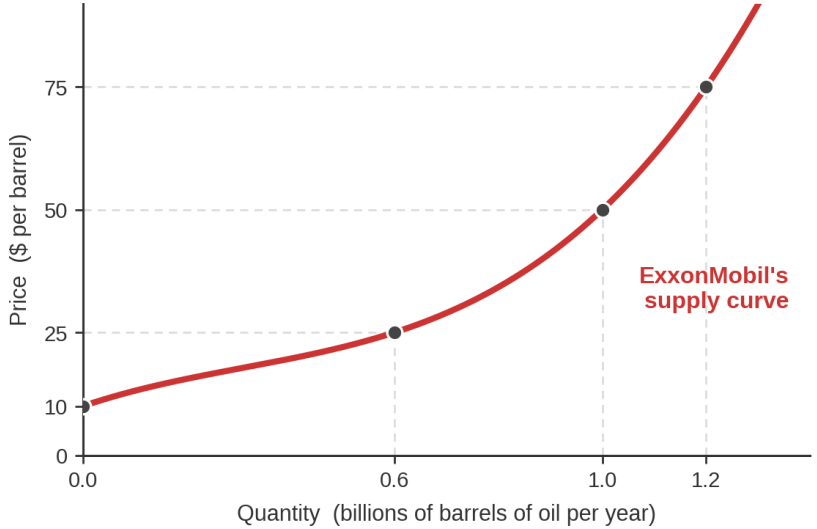


What would somebody have to pay you to hand this over?

Your quantity supplied at that price you have in mind is 1 — and below it, you supply zero. The price you probably hinges on how much it costs you to get a new one yourself.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Exhibit 4.6 — One Seller's Supply of Oil



EXXONMOBIL'S SUPPLY SCHEDULE

Price ($/barrel)	Quantity supplied (bn barrels/year)
$10	0.0
$25	0.6
$50	1.0
$75	1.2



At $10 a barrel it supplies nothing — no well it owns is worth operating.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Why Does It Slope Up?

Because producing more costs more per unit — so it takes a higher price to make the next unit worth producing.

THE CHEAP BARRELS GO FIRST

A firm pumps its easiest well before its hardest. The next barrel always comes from a more expensive source than the last one.

SO COST PER UNIT RISES

Overtime instead of regular hours. Older rigs. Deeper water.

More output, higher marginal cost.

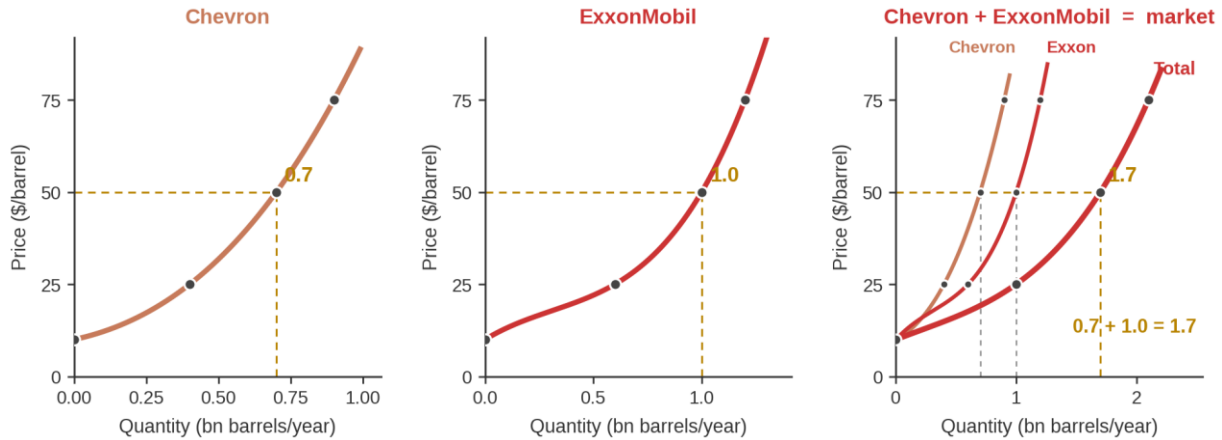
AND NEW SELLERS ARRIVE

In a market curve, a higher price also pulls in producers who could not cover their costs at the old price.

Same principle as the demand curve, mirrored: sellers keep producing while the price covers the marginal cost, and stop when it does not.

Exhibit 4.7 — Adding Sellers Together

Market supply is built exactly like market demand: fix the price, add the quantities across sellers.



at $50 a barrel: Chevron's 0.7 + ExxonMobil's 1.0 = the market's 1.7 billion barrels

Horizontal again. If your market curve came out steeper than the individual curves, you added the wrong way.

The Law of Supply

In almost all cases, the quantity supplied rises when the price rises ceteris paribus.

"Almost all cases" and "holding all else equal“ do the same work here as they did for demand.

Four Things That Shift Supply



INPUT PRICES

What it costs to make the thing. Labour, steel, fuel, rent.

Inputs cheaper → supply shifts RIGHT.



TECHNOLOGY

A better way to produce.

Always shifts supply RIGHT — you can make more for the same cost.



NUMBER OF SELLERS

More firms in the market → supply shifts RIGHT.

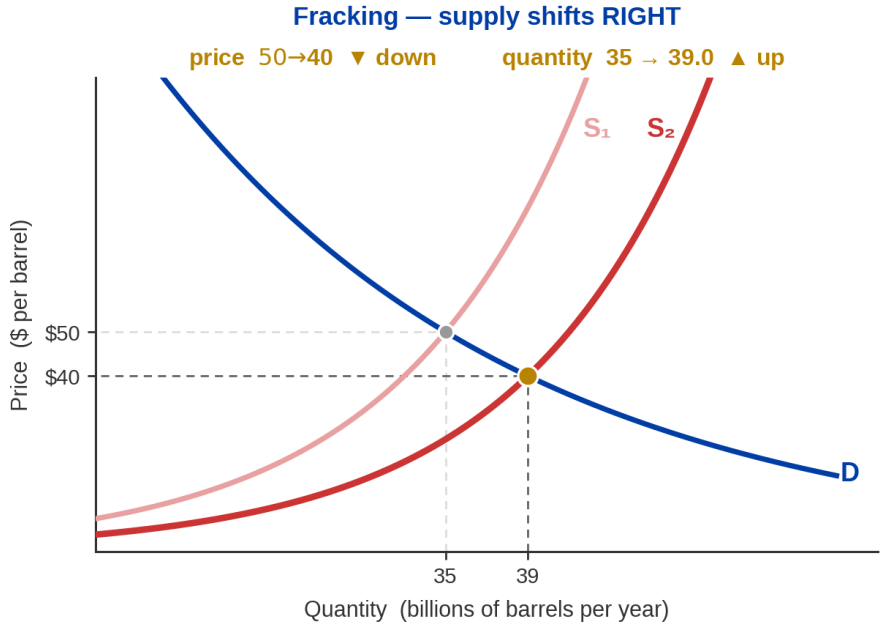
Firms exiting → LEFT.



SELLERS' EXPECTATIONS

Expect a higher price next month → store it, supply LESS now.

Technology: What Fracking Did to Oil Supply



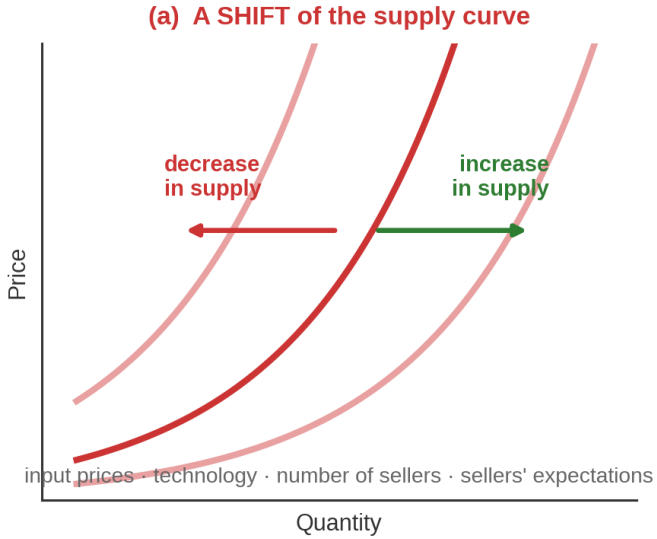
US CRUDE OIL PRODUCTION

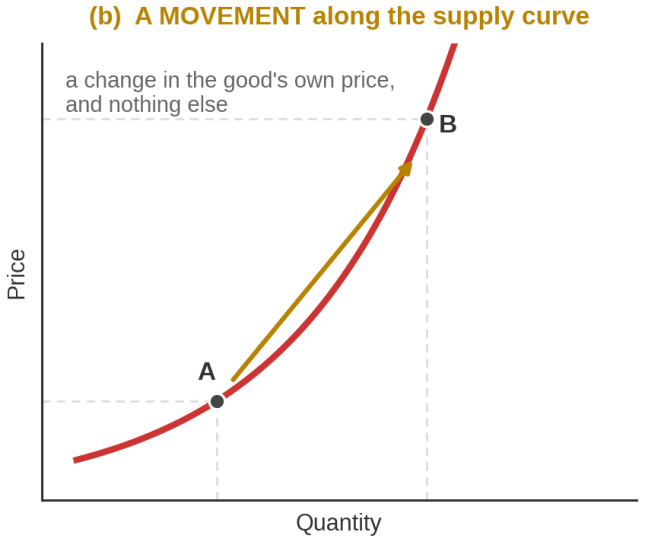
2008	5.0 million barrels a day
2013	7.5
2018	11.0
2025	13.6

A 2.7-fold increase. The US has been the world's largest crude producer every year since 2018.

US Energy Information Administration, field production of crude oil, annual. EIA, 9 July 2026. The curve shift above is drawn larger than life for legibility.

Shift vs Movement





Did the price of THIS GOOD change? Yes → movement along the curve. No → the whole curve shifts.

PART 4

Putting Them Together

Competitive Equilibrium

The **competitive equilibrium** price is the price at which quantity demanded equals quantity supplied. The quantity traded there is the equilibrium quantity.

NOBODY CHOOSES IT

Not the buyers, not the sellers, not a regulator. It is where two independent sets of plans happen to fit together.

IT IS A RESTING POINT

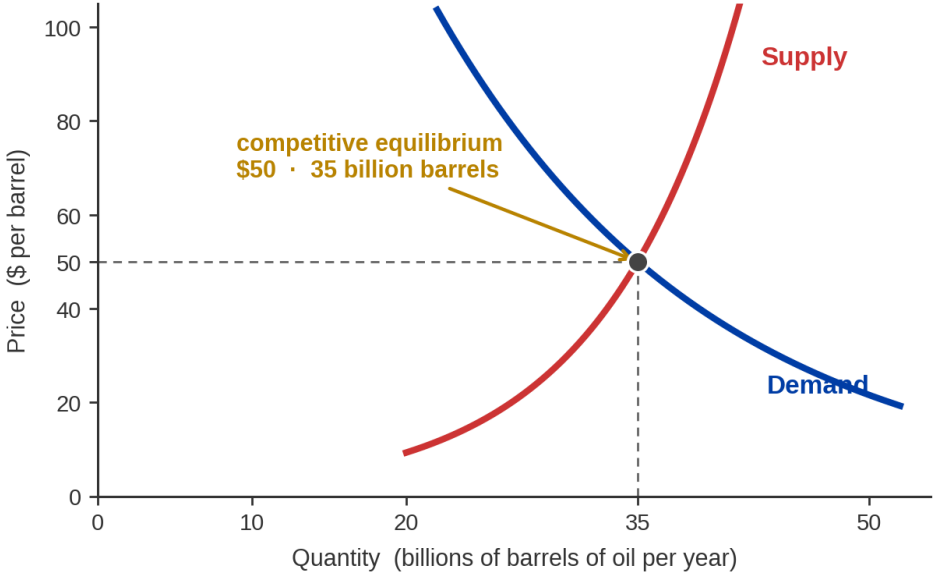
Chapter 1's word. At this price nobody has any reason to do anything differently.

Away from it, somebody does.



Equilibrium is the second of Chapter 1's three principles.

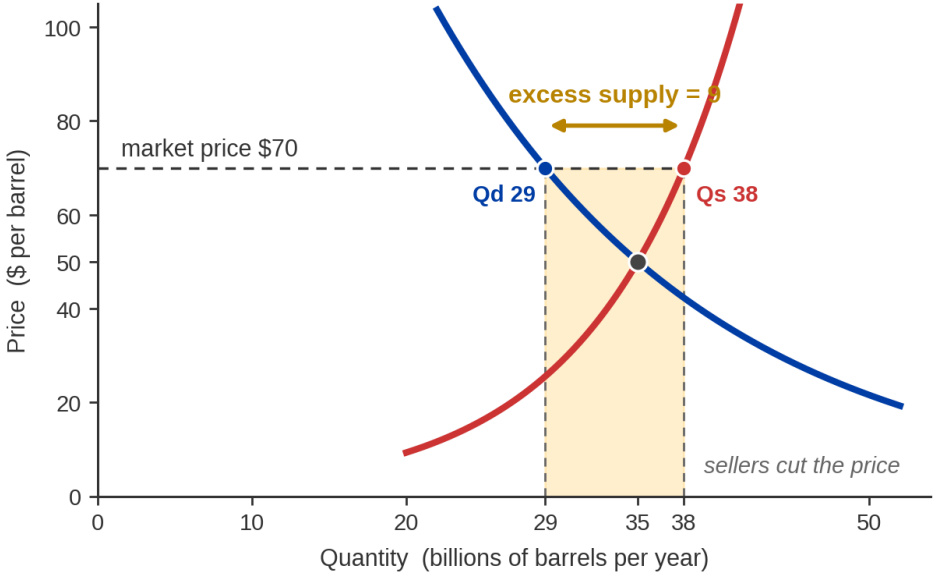
Exhibit 4.10 — Where the Two Curves Cross



READ IT IN THIS ORDER

1. Find the crossing point.
2. Go LEFT to the vertical axis — that is the equilibrium price. $50.
3. Go DOWN to the horizontal axis — that is the equilibrium quantity. 35 billion barrels.

Above Equilibrium: A Surplus



AT $70 A BARREL

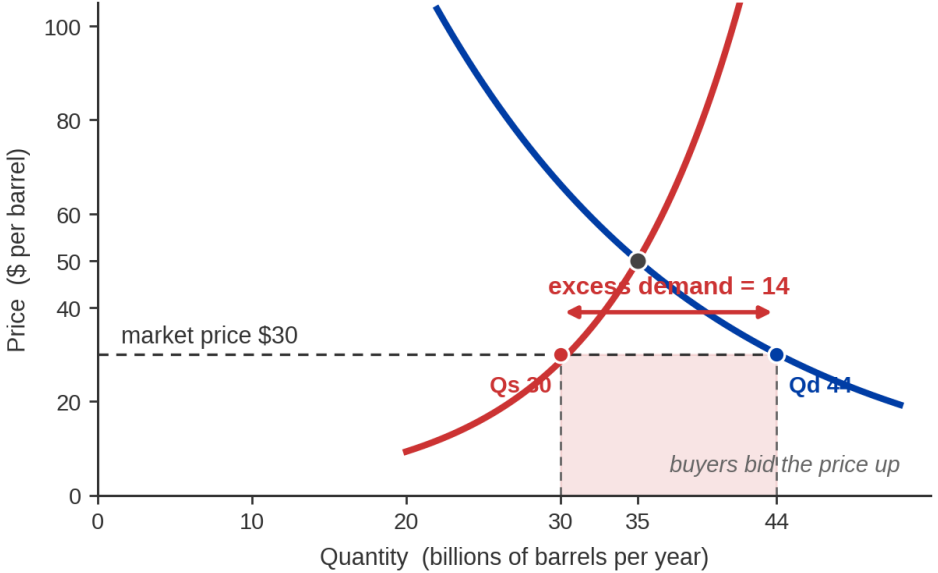
Quantity supplied 38
Quantity demanded 29

Excess supply 9

Nine billion barrels a year that sellers want to sell and nobody wants to buy.

Unsold oil is a signal, and sellers act on it: they cut the price. The cutting stops at $50, because that is where the surplus is gone.

Below Equilibrium: A Shortage



AT $30 A BARREL

Quantity demanded 44
Quantity supplied 30

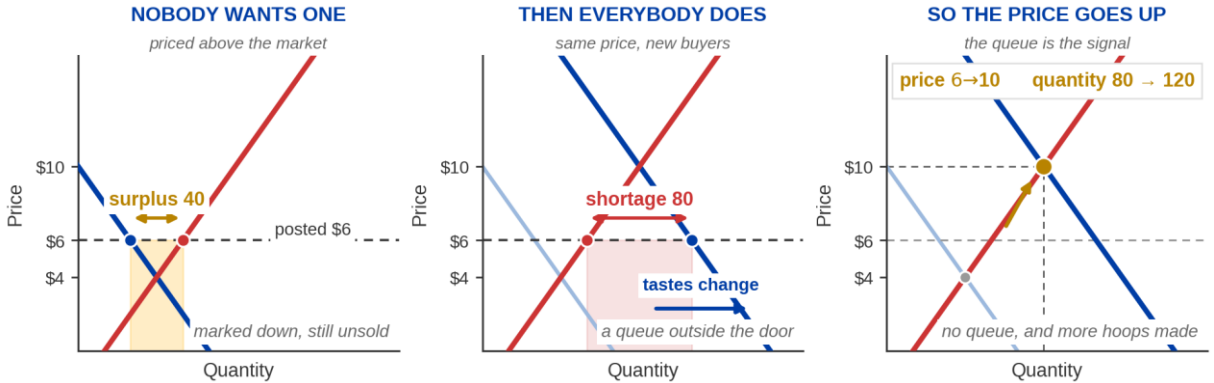
Excess demand 14

Fourteen billion barrels a year that buyers want and no seller will produce.

Buyers who cannot get any bid the price up. The bidding stops at $50 — the same place, arrived at from the other side.

The Same Product, Three Prices, Two Disequilibria

Nothing about the hoop changed. What changed was demand — and then, eventually, the price.



1 — PRICED ABOVE THE MARKET

At $6: 80 offered, 40 wanted. A surplus of 40, and marking it down does not shift them.

2 — THEN DEMAND SHIFTS RIGHT

Tastes change. Still $6: 160 wanted, 80 in existence. A shortage of 80.

3 — SO THE PRICE RISES

Buyers bid it up. It stops at $10, where 120 are wanted and 120 are made.

How to Work Any Shift Problem

Four questions, in this order, every time. Most lost marks come from skipping the first one.

01

WHICH CURVE MOVES?

Buyers affected → demand.
Sellers affected → supply.
If the good's own price changed, NEITHER — it is a movement, not a shift.

02

WHICH WAY?

More wanted, or easier to produce → RIGHT.
Less wanted, or harder to produce → LEFT.

03

WHERE IS THE NEW CROSSING?

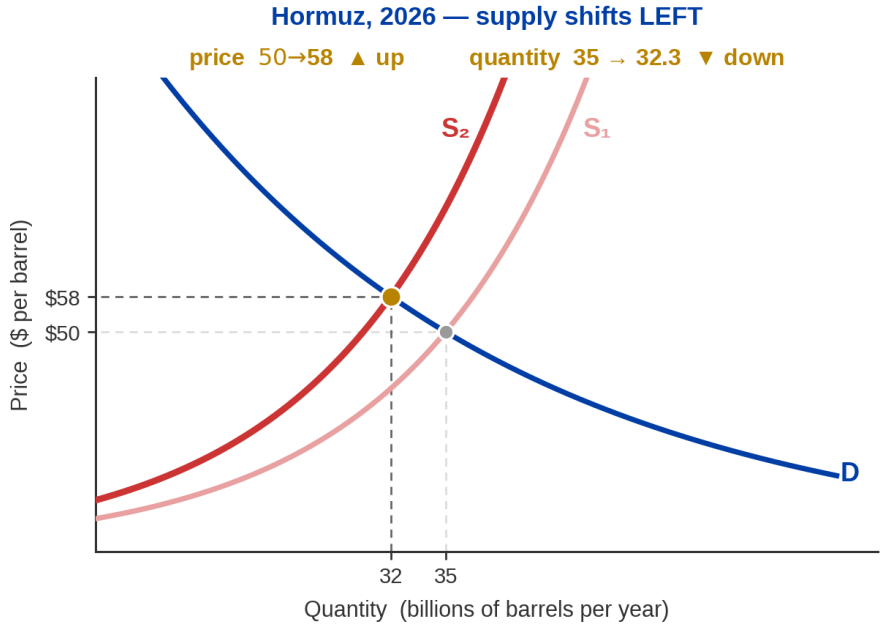
Draw the shifted curve. Find where it now cuts the curve that did not move.

04

READ OFF BOTH NUMBERS

Price: up or down?
Quantity: up or down?

Hormuz, 2026 — a Supply Shock in Progress



Hormuz Strait and Global Oil Production

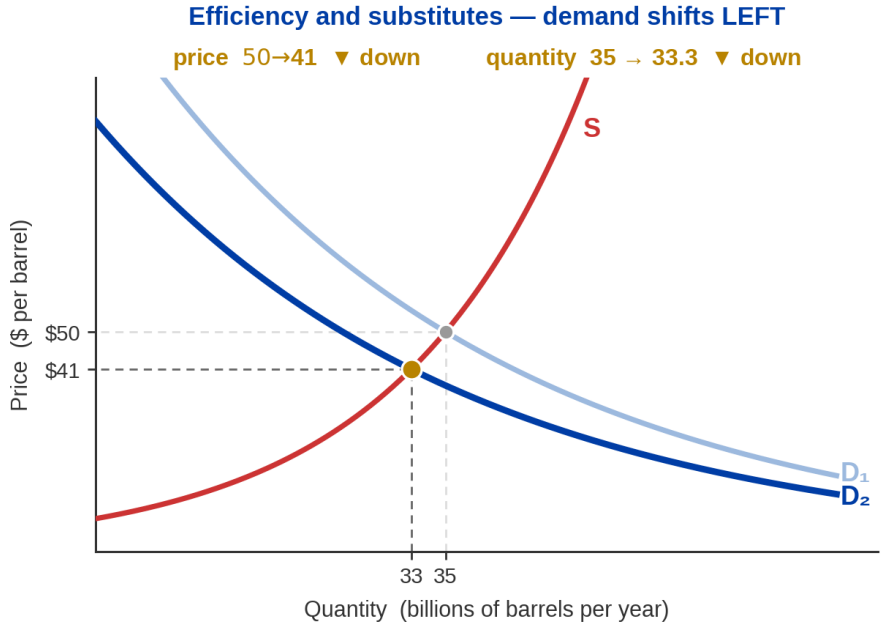
About 20 million barrels a day — 20% of the world's oil — moves through the Strait.



Flows fell from 20.7 to 14.6 million barrels a day between Q4 2025 and Q1 2026. Around 30% gone.

Strait volumes: International Energy Agency, February 2026. Quarterly flows: US Energy Information Administration Global Energy Security Data, 13 May 2026. Situation as of 3 August 2026 and moving — check before recording. The curve shift above is drawn larger than life for legibility.

Now Move the Other Curve



DEMAND FOR OIL FALLS

Some sources suggest the global demand for oil also slightly declined.

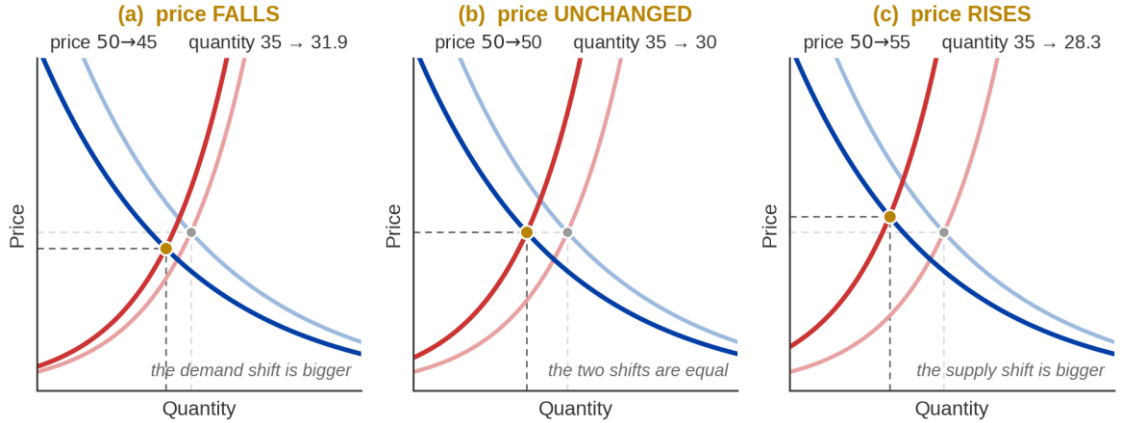


Better fuel economy. Electric vehicles. Efficiency standards. Environmental concern.

In this case, demand falls mainly in reaction to the event.

When Both Curves Move at Once

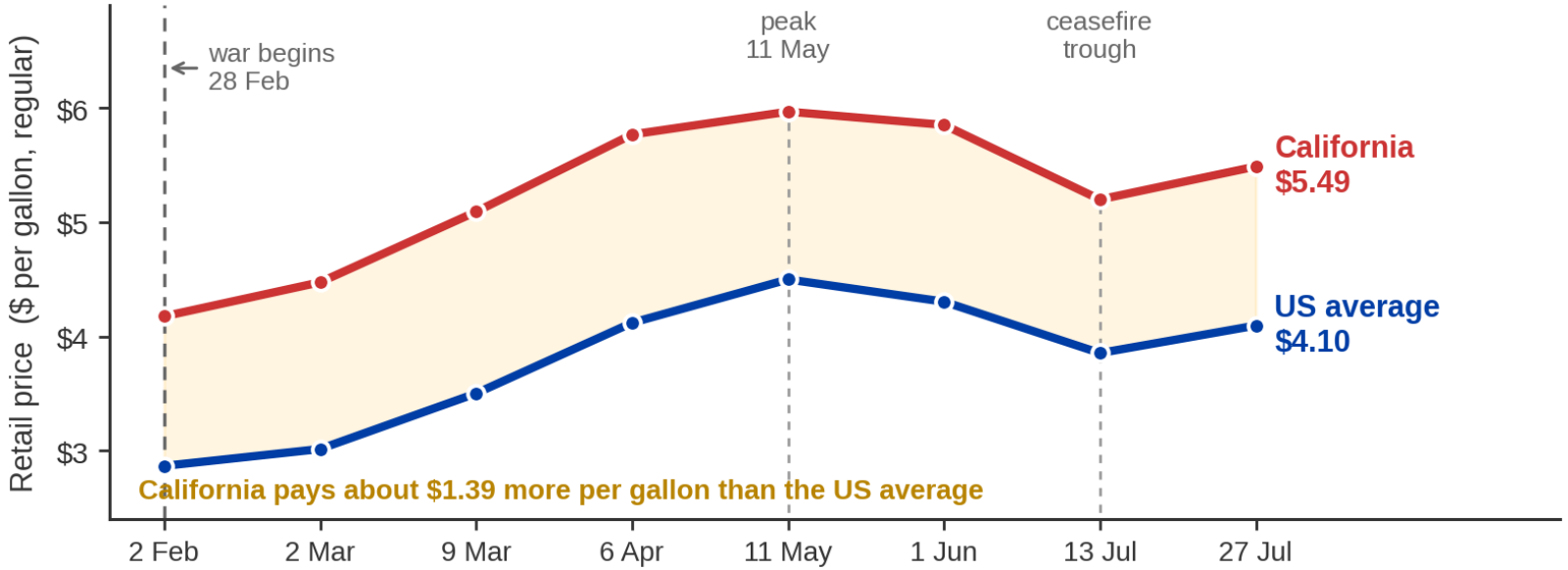
Supply shifts LEFT and demand shifts LEFT. Quantity definitely falls — and the price could do anything.



pale = before bold = after quantity falls in all three — only the price is left open

One direction is determined. The other depends on which shift is bigger — and the diagram alone cannot tell you.

Right Now: What Is Happening to Gas Prices

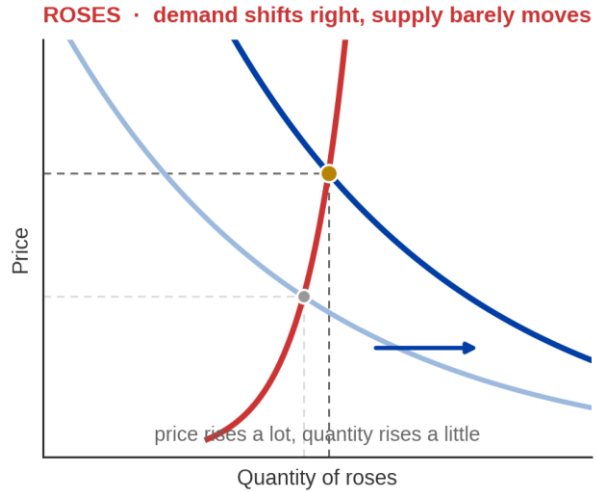


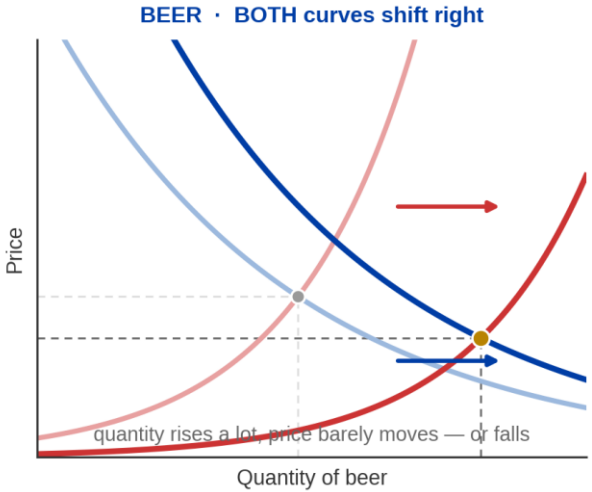
The US pump price rose 57% to a peak in May, gave back two-fifths of that by mid-July, and is climbing again.

US Energy Information Administration, weekly retail gasoline prices, all formulations, regular. Selected weeks. Prices as of the week of 27 July 2026 — re-check before recording.

Roses at Valentine's, Beer at a Holiday









ROSES — price roughly triples

USDA's weekly supermarket series: about $8 a dozen in August, close to $23 just before Valentine's Day.

BEER — price falls at peak demand

Chevalier, Kashyap & Rossi (AER 2003), 7.5 years of scanner data: beer margins DROP at Memorial Day, July 4th, Christmas.

Roses: USDA supermarket price series, reported by Jay Zagorsky, Boston University, February 2024. Beer: Chevalier, Kashyap & Rossi, “Why Don't Prices Rise During Periods of Peak Demand?”, American Economic Review 93(1), 2003.

PART 5

When Price Is Not Free to Move

1973–74: Capping the Price of Gasoline

A price ceiling is a legal maximum price. It only does anything if it is set BELOW the equilibrium price.

WHAT WAS CAPPED

Crude oil and refined products, including gasoline.

Controls began with Nixon's 1971 wage-and-price freeze —before the embargo — and were extended in 1973.

WHAT HAPPENED

Queues, station closures and state-level odd-even rationing by license plate.

In the last week of February 1974, about 20% of American gasoline stations had no fuel at all.

WHEN IT ENDED

Carter began phased decontrol in 1979.

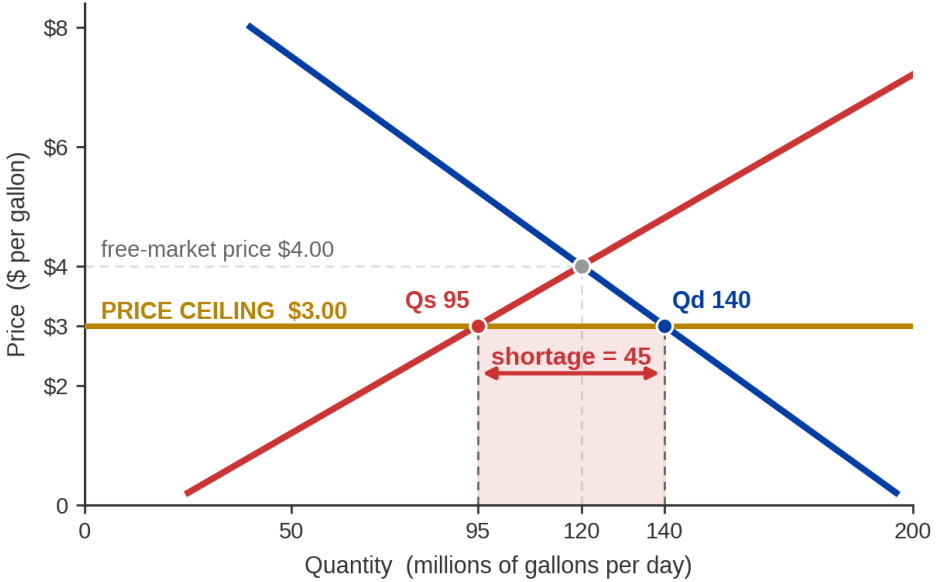
Reagan completed it in 1981, exempting all crude oil and refined products from price and allocation controls.



Not expensive gasoline. No gasoline.

Image: Acemoglu, Laibson & List, Microeconomics 3e

What a Ceiling Does to the Picture



AT A $3.00 CEILING

Free-market price	$4.00
Quantity traded	120
Ceiling	$3.00
Quantity demanded	140
Quantity supplied	95
Shortage	45

And only 95 gets traded — LESS than before the cap.

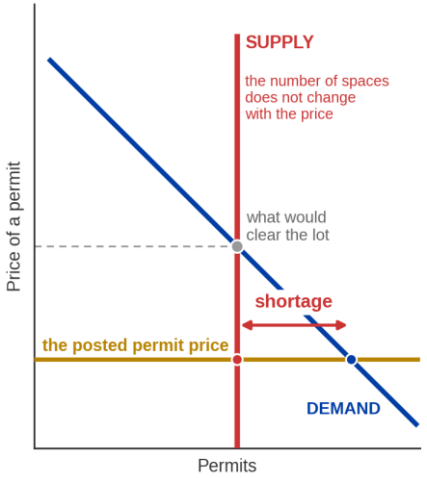
The cap was meant to make gasoline cheaper for buyers. It made it cheaper for the 95 who get some, and unavailable for the rest.

Campus Parking Lot

Why is parking on campus hard to find? Not because there are too many cars — because the permit price is held below the market-clearing level.



Lot 30: Lot Full on Wednesday afternoons



THE PRICE IS SET, NOT DISCOVERED

UCR posts permit rates rather than letting them clear. A Gold commuter permit runs on the order of $135–$145 a quarter depending on which rate sheet you read, and Gold Plus is described as available “in limited quantities.”

SO SOMETHING ELSE DOES THE RATIONING

Arriving at 8:30 a.m. instead of 10am. Circling the lot. Waitlists for specific lots.

The cost did not disappear — it stopped being money and started being time.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Practice: You Move, the Price Changes

2020 gasoline prices, calculated by Bloomberg. Assume your tastes and income do not change.

	Price per gallon	Your quantity demanded	Shift or movement?
United States	$2.40	your baseline	—
Kuwait (subsidised)	$1.27	RISES	movement ALONG your curve
Netherlands (heavily taxed)	$6.46	FALLS — lowest of the three	movement ALONG your curve
You take up rock climbing	unchanged	RISES at every price	SHIFT of your curve

Price changed → you slide along the curve. Anything else changed → the curve moves.

Chapter 4: The Full Picture

	What it says	Key terms	Where you saw it
Markets	Agents trading a good, plus the rules for trading it. Price is a selection device	market, market price, perfectly competitive	
Demand	Buyers want less as price rises, because the value of one more unit falls	quantity demanded, demand schedule, law of demand	Chloe's gasoline; the world's 35 billion barrels
Supply	Sellers offer more as price rises, because the cost of one more unit rises	quantity supplied, supply schedule, law of supply	ExxonMobil supplies nothing at $10 a barrel
Shift vs movement	Own price changes and you slide along the curve; anything else moves the whole curve	change in demand vs change in quantity demanded	Five demand shifters, four supply shifters
Equilibrium	The one price at which quantity demanded equals quantity supplied	competitive equilibrium, excess supply, excess demand	$50 and 35 billion barrels; Hormuz, fracking, roses
Price controls	Cap the price below equilibrium and the shortage cannot close — something else rations	price ceiling, shortage	Gasoline in 1974; parking on campus

Six rows. Two curves, one crossing, and a list of the things that move each of them.

KEY TAKEAWAYS

Chapter 4: Demand, Supply, and Equilibrium

- A market is agents, a good, and rules for trading — no address required
- Demand slopes down and supply slopes up, both for the marginal reasons from Chapter 3
- Own price changes and you move along the curve; anything else shifts the whole curve
- Equilibrium is where the curves cross — above it price falls, below it price rises
- Freeze the price and the shortage cannot close, so something else rations, usually a queue